

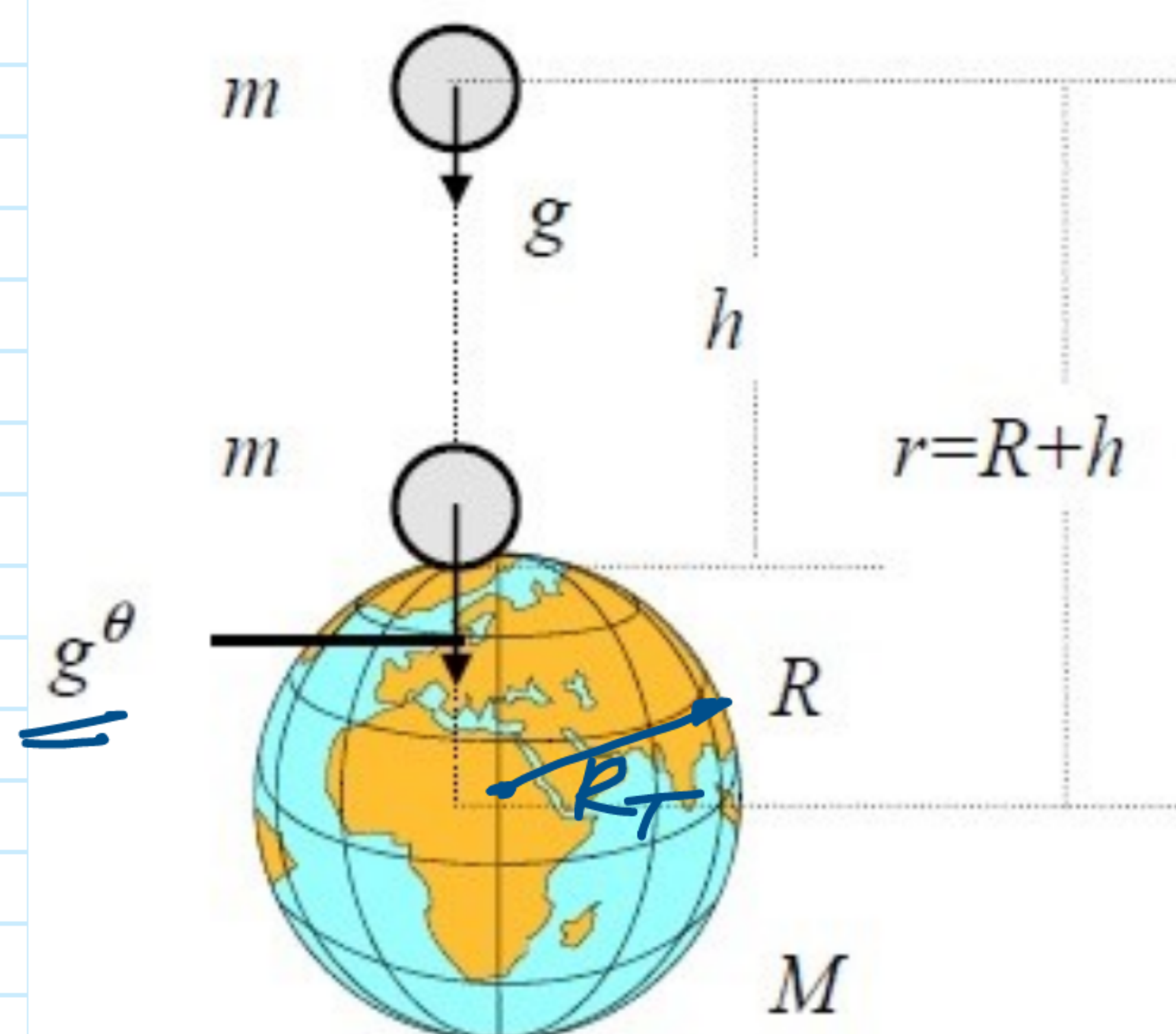
Ligjerata 7c

Vartësia e nxitimit të rendimit tokësor nga lartësia

$$F = \gamma \frac{m_1 m_2}{r^2} \quad \begin{matrix} m_1 = m \\ m_2 = M_T \end{matrix}$$

$$\left. \begin{aligned} F &= \gamma \frac{m M_T}{R_T^2} ; r = R_T \\ F &= Q = mg^0 \end{aligned} \right\} \textcircled{1}$$

$$\left. \begin{aligned} F_h &= \gamma \frac{m M_T}{(R_T + h)^2} ; r = R_T + h \\ Q &= mg \end{aligned} \right\} \textcircled{2}$$



$$\textcircled{1} \quad \cancel{m} g^0 = \gamma \frac{\cancel{m} M_T}{R_T^2} \Rightarrow \boxed{g^0 = \gamma \frac{M_T}{R_T^2}}$$

$$\textcircled{2} \quad \cancel{m} g = \gamma \frac{\cancel{m} M_T}{(R_T + h)^2} \Rightarrow g = \gamma \frac{M_T}{(R_T + h)^2}$$

$$\frac{g}{g^0} = \frac{\gamma \frac{M_T}{(R_T + h)^2}}{\gamma \frac{M_T}{R_T^2}} = \frac{R_T^2}{(R_T + h)^2} \Rightarrow g = g^0 \left(\frac{R_T}{R_T + h} \right)^2$$

$$g^0 = \gamma \frac{M_T}{R_T^2} \quad / \cdot \frac{R_T^2}{\gamma} \rightarrow \boxed{M_T = \frac{g^0 R_T^2}{\gamma}}$$

$$g^0 \approx 9.81 \frac{m}{s^2}$$

$$R_T = 6378 km$$

$$\gamma = 6.67 \cdot 10^{-11} \frac{Nm^2}{kg^2}$$

$$M_T = \frac{9.81 \frac{m}{s^2} \cdot (6.378 \cdot 10^6 m)^2}{6.67 \cdot 10^{-11} \frac{Nm^2}{kg^2}}$$

$$\boxed{M_T = 5.9 \cdot 10^{30} kg}$$